

VPS72 p.Lys115Arg | JLF V3 | 27 aa

disclosed minimal epitope ERALLPLEL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

0.5 display

Pepsickle human (in vivo) native 0-1

0.5 display

NetCleave I native 0-1

0.5 display

NetChop 20S (in vitro) native 0-1

0.5 display

MHC-I: 5 shown / 12 eligible (top-10 cap)

C01:02 0.31%

C07:01 0.25%

B27:05 1.1%

C07:01 1.7%

B08:01 0.89%

G | S | S | Q | K | A | R | E | E | R | A | L | L | P | L | E | L | Q | D | D | G | S | D | S | R | K | S

1

5

10

15

DR03:01 3.1%

25

27

MHC-II: 5 shown / 10 eligible (top-10 cap)

DR03:01 2.7%

DR03:01 2.2%

DR03:01 3.1%

DR03:01 2.3%

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

G | S | S | Q | K | A | R | E | E | R | A | L | L | P | L | E | L | Q | D | D | G | S | D | S | R | K | S

1

5

10

15

20

25

27

DPP4 / qPISA

N-terminal 2-residue removal

21% predicted loss

MME / neprilysin

Internal hydrophobic preference

FAP - internal

Gly-Pro|non-Pro recognition

FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

ANPEP / CD13

exposed N-terminus only

ENPEP

exposed N-terminus only

Triangle = motif match | hollow circle = no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted peptide-signal loss in the source 4 h assay. Native scores remain in CSV. Motifs indicate sequence matches only.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.